

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

Why Software Tests Fail and Hardware Attestation Wins

Based on: [arXiv:2504.04715](https://arxiv.org/abs/2504.04715)

For ML Researchers | AI Security Practitioners | Cloud Infrastructure Engineers

The Model Substitution Problem

Economic Incentives Drive Covert Swaps

Commercial LLM API providers charge for access to specific models, but the black-box nature of APIs means users have no guarantee the advertised model is actually being served.

Providers face strong economic incentives to covertly substitute cheaper alternatives to reduce GPU costs while maintaining pricing.

Formalized as Hypothesis Testing

H_0 : API serves $P_{\text{spec}}(y|x)$ – the advertised model

H_1 : API serves $P_{\text{actual}}(y|x)$ – a cheaper substitute

Can an auditor distinguish these distributions from samples alone?

SUBSTITUTION TYPES

1

Smaller Model

8B instead of 70B

2

Quantized Variant

FP8/INT8 instead of FP16

3

Fine-tuned / Updated

Modified weights, same name

4

Different Family

Entirely different architecture

FP8 Quantization Preserves Most Accuracy

Making Substitution Hard to Detect

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

- For most models, FP8 scores fall within the error bars of the original—especially on MMLU and GSM8K—making statistical detection extremely challenging.

Higher Temperature Amplifies Performance Gaps

but Increases Noise

Key Observations

Temperature 0.2 – 1.0

Original and FP8 models perform similarly; the substitution signal is weak or indistinguishable.

Temperature 2.0

All models degrade significantly. The gap between original and substitute narrows or becomes noisy, further complicating statistical detection.

Implication

Even when temperature scaling amplifies differences, the increased variance makes reliable statistical separation impractical.

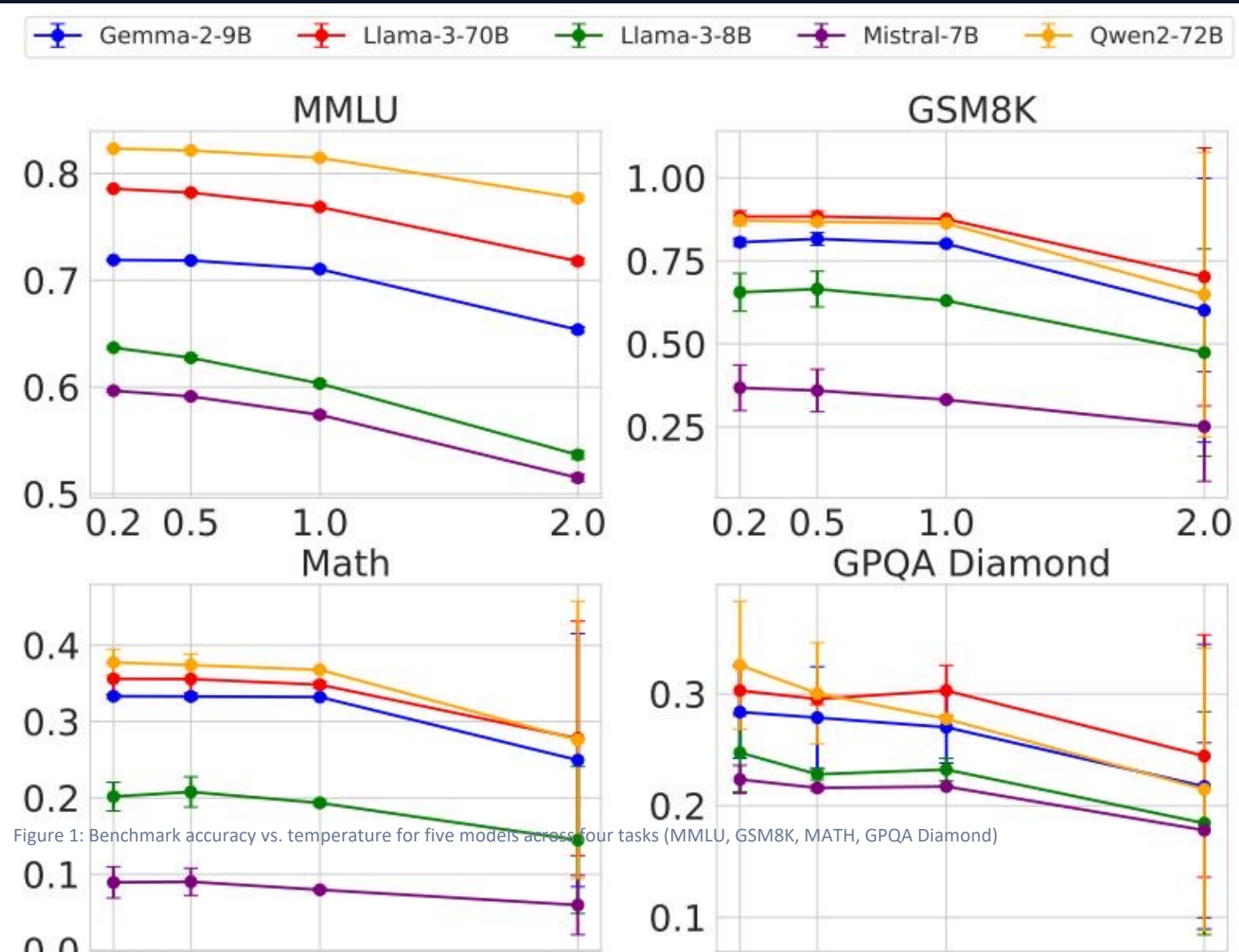


Figure 1: Benchmark accuracy vs. temperature for five models across four tasks (MMLU, GSM8K, MATH, GPQA Diamond)

Log Probability Fingerprinting Fails

Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Why Fingerprinting Breaks

Same model, different outputs

Llama-3-8B produces different log probabilities depending on inference software and hardware.

Software variation

transformers 4.40 vs. 4.50, vLLM 0.6.2 vs. 0.8.2 — all introduce log-prob shifts.

Hardware variation

A100 vs. H100 GPUs produce measurably different probability distributions for the same model.

Result

An auditor cannot distinguish legitimate infrastructure variation from actual model substitution.

Figure 5: Log probability variation across inference frameworks and hardware for Llama-3-8B

Randomized Routing at Low Rates

Makes Statistical Detection Query-Prohibitive

The provider mixes substitute responses with genuine ones at probability p :

$$P_{\text{mix}}(y|x) = (1 - p) \cdot P_{\text{spec}}(y|x) + p \cdot P_{\text{sub}}(y|x)$$

As $p \rightarrow 0$, $P_{\text{mix}} \rightarrow P_{\text{spec}}$ — the mixed distribution converges to the specified model

The Adversarial Strategy

- Provider routes fraction p of queries to the cheaper model
- Even at $p = 0.1$, the provider saves 10% of compute costs
- Detecting $p \approx 0$ requires exponentially more queries
- Sample complexity scales as $O(1/p^2)$ for reliable detection

⚠ BENCHMARK EVASION

Providers can also detect known benchmark prompts and route them to the genuine model while serving substitutes for regular traffic.

Combined Attack

Randomized substitution + benchmark evasion renders all software-only detection fundamentally query-prohibitive.

Bottom Line

No practical number of API queries can reliably detect a sufficiently low-rate substitution under adversarial conditions.

Software-Only Auditing Methods

Are Fundamentally Unreliable

Method Type	Weakness	Practical Limitation	Verdict
Text-Based Statistical Tests (e.g., MMD)	Requires impractically large query budgets; fails against subtle or randomized substitutions	FP8 quantization shifts scores within error bars—no statistical separation achievable	UNRELIABLE
Log-Probability Fingerprinting	Defeated by production-level inference nondeterminism across software versions and GPU hardware	Cannot distinguish infrastructure variation from actual substitution; framework and GPU differences dominate	UNRELIABLE
Adversarial Countermeasures	Randomized routing and benchmark evasion further undermine all detection approaches	Low substitution rates require $O(1/p^2)$ queries; benchmark prompts can be selectively routed	FATAL

Conclusion: No software-only method can provide reliable model integrity verification under realistic adversarial conditions. Each approach has fundamental limitations that cannot be overcome through more queries or better algorithms alone.

Trusted Execution Environments

Provable Cryptographic Model Integrity

How TEEs Solve the Verification Problem

Hardware-secured enclaves provide cryptographic attestation of the code and data being executed inside them.

What TEEs Guarantee:

- ✓ Correct model weights are loaded and executed
- ✓ Inference code runs unmodified inside the enclave
- ✓ Attestation reports are cryptographically signed by hardware
- ✓ Only modest performance overhead vs. untrusted execution

Practical Deployment Path

NVIDIA Confidential Computing on H100 GPUs enables TEE-based inference today. Unlike Zero-Knowledge Proofs (ZKPs), which are computationally prohibitive for large models, TEEs are practical and deployable at production scale.

TEE vs. Software Methods		
Property	Software-Only	TEE
Verifiable	No	Yes
Efficient	Query-heavy	Modest overhead
Adversary-robust	Defeated	Provable

Key Takeaways

Close the Verification Gap with Hardware Attestation

1

Economically Motivated, Practically Undetectable

Model substitution is driven by GPU cost savings and cannot be reliably detected through software-based methods alone.

2

FP8 Quantization Barely Affects Benchmarks

FP8 scores fall within error bars of original models on MMLU and GSM8K, making quantization-based substitution nearly undetectable.

3

Inference Nondeterminism Kills Log-Prob Fingerprinting

Software framework versions and GPU hardware introduce log-probability variations that cannot be distinguished from actual substitution.

4

Randomized Substitution Defeats Statistical Tests

Low-rate mixing and benchmark evasion make detection query-prohibitive, requiring exponentially more samples as $p \rightarrow 0$.

5

TEEs Are the Only Viable Path to Verifiable LLM APIs

Hardware attestation provides provable, efficient, and robust model integrity guarantees—deployable today via NVIDIA confidential computing.

Call to Action: Adopt hardware-attested inference pipelines to guarantee model integrity in commercial LLM APIs.